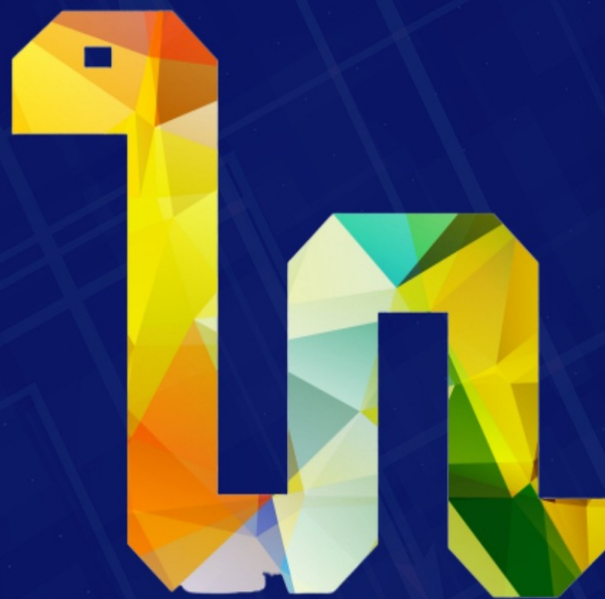


PYTHON

— PROGRAMMING —

THE ULTIMATE CRASH COURSE

for Learning Python Quickly, with Practical Examples and
Coding Language Tricks for Beginners



Daniel Howard

2019
edition

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*Computer programming for data
science and machine learning.*

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Introduction

Congratulations on purchasing *Python Programming*, and thank you for doing so.

The following chapters will discuss many of the different topics that you need to know when it is time to learn how to work with the Python programming language. There are a lot of companies and individuals who want to be able to learn how to code, and the Python language is going to be one of the best options for you to choose to go with. This guidebook will make sure that you are able to get some of the basics of this process down, so you can write your own codes down in no time, no matter what you would like to use these for.

To start, this guidebook is going to take a look at some of the basics that come with Python. We are going to spend some time looking at the benefits of working with Python and why it is such a good language to work with, and then we will spend some time exploring what this language takes to get installed on the various different operating systems that are out there. From there, we can look at a few of the basic parts that come with this code, including the objects, comments, keywords, functions, and more.

Now that some of those basics are out of the way, it is time for us to take a look at some of the coding that we are able to do when it is time to work with this language. The best way to learn anything about coding and how to make this happen, we first need to look at some of the codes and get some practice with those. Here we are going to learn how to work with the different data types that are there, how to work with the functions, the importance of classes in the Python language, and how to write out some of our own conditional statements. All of these are going to provide us with some really great codes that we are able to write with the help of the Python language.

After we have had some time to learn those different parts of the coding world, it is time to look at some of the tips and tricks that we are able to do when it comes to learning how to code. No matter how much we prepare, there are times when things are not going to work the way that you want. These tips are going to show you some of the things that you can do when

your code is not behaving the way that it should.

The end of this guidebook is going to spend some time looking at the basics of data analysis and why this is an important part of the process. Many companies are jumping on board and looking at how to work with data science and data analysis easier than ever before. You will quickly be able to keep up with the benefits of working with this, the different steps that you need to take in order to work with a data analysis for your own projects, and some of the best algorithms that we are going to be able to work with when it comes to the Python language and data analysis.

There is so much that we are able to work in order to get started with the Python language, and this guidebook will spend some time talking about the most important parts of this process so that you are able to make it work for all of your business need. When you are ready to learn more about Python programming and how you can work with this on data science or a data analysis project, make sure to check out this guidebook to get started!

Every effort was made to ensure it is full of as much useful information as possible; please enjoy it!

Chapter 1: What is the Python Language, and Why Should I Use It?



There are a lot of different coding languages out there that we are able to work with. We may find that some of these are going to work the best when they help us create webpages and work online. Others are going to be able to handle some of the more technical aspects of our work, whether we are working with math or science, and can tell us a lot about the world, the trends around us, and more. And still, others are able to be there to help a beginner get into the world of coding.

If you are just getting started with the idea of coding, then the best option that you can choose to learn how to code is with the Python language. This language is simple to learn while having all of the features and power that you need to get coding done. This language is going to come with the standard library and plenty of extensions to get the work completed. And as we go through this guidebook, you will quickly see why this language is

actually pretty easy for you to learn, even as a beginner.

There are a lot of programmers out there who love to work with the Python language and wouldn't choose anything else to work with for their coding needs. And this could easily become you with a little bit of knowledge and practice along the way. When you are ready to learn a bit more about the Python language and how it works, make sure to check out the rest of this guidebook to see what we are talking about!

How to Use Python

While we are here, we need to spend some time taking a look at some of the major uses that come with the Python language. Many companies are more than willing to learn how to work with the Python language for some of their big computer tasks. You will find that Python is already making a bit different in some of the methods that companies use to work well, and towards making sure that the bottom line is not going to bottom out in the process. Here are some of the different ways that we are able to work with the Python language in order to see just all of the capabilities that can come with this language.

The first way that we are able to work with Python is to help us to develop some applications. This can include things like a software development application, network programming, games, web applications, GUI based applications, and more. This is due to the fact that Python is able to make an interface that is interactive and is easy to develop applications with.

Python is also going to be able to help out with multiple programming paradigms. Python is going to provide us with some of the support that is needed to handle these. As it is able to support OOP, as well as support for structured programming, Python is going to come with a bunch of features that allow it to support the concepts and more that comes with a functional programming language. We are able to use this language to help out with a system that is dynamic, and even with automatic management of the memory.

We will also see that Python is going to have a robust and fairly large library that is going to be useful for developing any applications that you would like.

It is also going to make it easier for a developer to rely more on Python compared to the other languages as well. The standard Python library is going to help us use a different range of modules that are available so that we can complete any of the tasks that you would like.

Next on the list is that the Python language is going to be compatible with a lot of major systems and platforms. This is going to make it easier for us to complete a lot more projects because we will be able to finish them no matter what platform or system we are working with. Python is going to be so compatible because it is used mainly for developing some new applications.

With the help of the interpreter, the code is going to be run on the right platform, and it has the tools that are needed to support most of the major operating systems that are out there. This makes it easier to pull up the programs and more that come with Python and get them to work no matter what you plan to do with them or where they need to work.

If you plan to do some work with databases while you are doing your coding, then Python is going to be a good one to use to help with this. You will find that the Python language is going to make it easier for us to access the databases that we would like. Python is going to help us to customize the interfaces that come with most of the common databases, making it easier for us to get some of the work done that we would like in the process.

These, of course, are just a few of the ways that you are able to work with the Python language and see some results in the process. When you are ready to handle some coding, no matter what kind of coding you are looking to work with, you will find that the Python language is going to be one of the best choices to go with. Python is going to have all of the tools and the features that you are looking for, and it is easy to use as well. This all comes together and ensures that you are going to be able to finish any project that you would like with the help of this coding language.

The Benefits of Python

When it is time to start working with a coding language, you will find that there is no one out there that is better than the Python coding language. There are so many benefits that you will be able to receive when it is time to work with Python that it will not take long before you are able to see why it is the one that most programmers are going to choose for their needs. Let's take a look at some of the benefits that you will be able to get when you work with this kind of coding language.

The first benefit is that this language is designed to be easy to read and write. Unlike some of the options that have been around for a long time and are more traditional coding languages, Python wanted to be easy for even beginners to learn how to use. Whether you have been a coder in the past and have some experience with making all of this work, or this is the first time that you have ever done anything with coding, you will find that the Python language is the best choice to go with.

Even though there is a nice learning curve when it is time to work with the Python language, you will find that this is not going to take away any of the functionality or the capabilities that you will see with this language. Python is a language that has power, and it often has more power than those who are first exploring it will think. In addition to doing some basic coding that you would like and that we will explore more in this guidebook, Python comes with the extensions and more that will help you handle data analysis, data science, machine learning, and more.

The large community that comes with Python is another added bonus that you are likely to enjoy as well. This is a community of all kinds of programmers found all throughout the world. You will be able to work with beginners all the way up to advance, with those who are just beginning with the work of coding for the very first time, all the way to professionals who have been doing it for a very long time.

This is a great thing for someone who is just getting started with Python for the first time. This allows you to find a bunch of other programmers, many with some experience in coding, who can answer your questions, show you how to try something new with the coding that you want to do, and more.

Any time that you get stuck or want to learn something new with Python, this community will be able to help you out.

Python is also known as an OOP or object-oriented programming language. This is good news for someone just starting because it has a nice logical way that you are able to work with organizing your code. You will find that this is going to make it easier for you to store all of your objects, or parts of the code, into the right class, and then everything will be right where you left it when it is all done. This will make more sense when we explore it a little bit later, but it is basically going to make things easier when it is time to write the code.

All of the libraries that come with this kind of language are going to be nice to work with as well. The traditional or standard library that comes with Python is able to take on a lot of different tasks and can help with some of the basic codings that you want to do. But if you would like to do something that is a little more complex, such as what we see with data analysis and machine learning, then you may need to add something on to Python to make this work.

The good news is that Python comes with the capabilities and extensions that you need to make all of this happen. There are a ton of data science and machine learning libraries out there that match up perfectly with the Python coding language. This makes it easier for you to work on creating the kinds of codes that you need with Python, no matter how simple or how complex they may be.

As we can see, there are a lot of different parts that come with the Python language, and understanding what these parts are and how they work together is going to be so important when it is time to work on some of your codes. With this in mind, let's take some of those basics and explore some of the neat things that you are able to do with the Python language.

Chapter 2: How Can I Install Python on My Computer?



Now that we have had a chance to talk about Python a little bit, and some of the reasons why you would want to choose to work with this coding language rather than one of the others that are out there, it is time to get to work. Before we are able to jump in and start doing some of the codings that need to be done, we first need to take some time to learn the proper steps needed in order to install Python on your system

The good news, as we mentioned a little bit in the past chapter, is that Python is able to work with any kind of operating system or platform that you would like to focus on. This is good news because it ensures that we are able to go through and work with this language without changing up our system, no matter what system is the one that we are using, or the one that we will want to work with for this project.

The first thing that we need to consider here is which version of Python we would like to work with. There are a few different options here, which is going to ensure that we are going to be able to pick the version that is right

for you. Python is an open-sourced language, which means that it is free to install and use, and can be modified to fit your needs. There is still a development team that works on fixing the bugs and making the updates that are necessary. This is good news for you, but it does mean that you have to figure out which version is the right one for you.

There are Python 2 and Python 3 right now, with a few updates that fit into each one. They are similar, but they will often depend on the kinds of features that you would like to use. If your computer is already providing you with a version of Python 2, then this is a good one to work with, and you don't have to update it if you do not want to in this process. It is going to be pretty similar to Python 3 and some of the codes that we will do in this guidebook.

However, most programmers are going to work with Python 3. This is the newest version of the Python language that is out there, and it is going to have some of the best features and more that you need for coding. And it is likely that at some point, the developers who work on Python 2 will stop providing support for that version, so going with the newer option is going to be the best option to go with.

You will be able to find both of these downloads online to work with overall, so you have to choose which one is the best for you. You can also find versions of these that will work on all of the main operating systems so it should not be too hard to add these onto whichever system you are working with.

Another thing to keep in mind here is that with the coding we are going to look at below, and with the steps that we will look at in order to install Python on all three of the major operating systems, is that we are going to get our files from www.python.org . Python is going to come with a compiler, a text editor, and an IDE, along with some other files as well. These are all important to ensure that the program is going to run properly, and if one is missing, then you will not be able to get any of the codes that you want to work on.

There may be some other third-party sights that you are able to give a look at and then download those instead. This is an option, and depending on what

you would like to do with the Python language when you are done, you may want to do this. But not all of them are going to provide you with all of the extras that are needed. You will have to go through and download each of the parts to make sure that you are not missing something.

On the other hand, when you download from the link that is provided at www.python.org, you will be able to know for a certainty that all of the files and parts that you need are there. you just need to go to the link there, click on the operating system that you are working with, and then move on to let it download on your system without any problems along the way at all.

Now that we have had some time to take a look at some of the things about installing Python on your computer, no matter which operating system you will use, it is time to take a look at the specific steps that are needed to make this happen.

Installing Python on Mac OS X

The first option that we are going to take a look at when it is time to install Python on our computers is the Mac OS X. If you are working off a computer that will rely on the Mac operating system, then you should notice early on that there will be some kind of version of Python 2 already on the computer. The exact version is something that can change based on how long you have had your computer, but we can easily check on which version by opening up our terminal app and work with the prompt below to get it done:

Python – V

This is going to show you the version you get so a number will come up. You can also choose to install Python 3 on this system if you would like, and it isn't required to uninstall the 2.X version on the computer. To check for the 3.X installation, you just need to open up the terminal app and then type in the following prompt:

Python3 – V

The default on OS X is that Python 3 is not going to be installed at all. If you

want to use Python 3, you can install it using some of the installers that are on Python.org. This is a good place to go because it will install everything that you need to write and execute your codes with Python. It will have the Python shell, the IDLE development tools, and the interpreter. Unlike what happens with Python 2.X, these tools are installed as a standard application in the Applications folder.

You have to make sure that you are able to run both the Python IDLE and the Python shell on any computer that you are using. And being able to run these can be dependent on which version you have chosen for your system as well. You will need to go through and check on your system which version of Python is there before proceeding and then moves on from there to determine which IDLE and shell you need to work with.

Installing Python on a Windows System

The second option that we are going to take a look at when it is time to work with the Python language on an operating system is on a Windows computer. There are many people who rely on the Windows operating system, so it is important to know how to go through this process and how to get Python set up. This process is going to take a bit longer than the other two, mainly because Windows already has its own coding language pre-installed, so we have to take some of the extra steps that are needed to get this done.

The good news is that even though Python is going to not be the automatic option on your computer doesn't mean that it won't work great. You will find that this coding language will be fine on Windows, and there will be a lot that you are able to do with it on these systems. It is just there to give you some notice that you have to take on a few extra steps.

Now that we have that out of the way, some of the steps that you will need to follow in order to make sure that the Python language is set up and ready to go will include the following:

1. To set this up, you need to visit the official Python download page and grab the Windows installer. You can choose to do the latest version of Python 3, or go with another option. By default, the

installer is going to provide you with the 32-bit version of Python, but you can choose to switch this to the 64-bit version if you wish. The 32-bit is often best to make sure that there aren't any compatibility issues with the older packages, but you can experiment if you wish.

2. Now right-click on the installer and select "Run as Administrator." There are going to be two options to choose from. You will want to pick out "Customize Installation."
3. On the following screen, make sure all of the boxes under "Optional Features" are clicked and then click to move on.
4. When you are still under the part for Advanced Options, you need to spend a few moments figuring out where you would like the Python files to be installed. You can then click on the Install button. This will take a few minutes to finish, so have some patience and wait. Then, when the installation is done, you will be able to close out of it.
5. After this part is done, it is time for us to go through and create the PATH variable that we need for the system. We want to have it set up so that it includes the directories and all of the packages and other components that are needed later on. The steps that we need to use to make all of this fit together will include:
 - a. First, we need to get the Control Panel opened up. We are going to do this by clicking on the taskbar and then typing in the Control Panel. Then we are going to click on the icon.
 - b. Inside of this same Control Panel, do a search for the table of Environment. Then we can click on the button for Edit the System Environment Variables. When we are on that part, we are able to click on the part that says Environment Variables.
 - c. Inside of the Environment Variables, we are going to find a tab for User Variables. You get two choices here of either editing the PATH variable that you find there, or you can create a brand new one.
 - d. If there isn't a variable for the PATH already on the system you are using; then it is time for us to go through

and create our own. You can do this by clicking on the button for New, make a name for the variable of that PATH, and then add in the directories that you would like. Then it is time to click or close out of all the dialogs that have come up for the Control Panel and go to the next step.

6. When you reach this part, you should be able to open up the Command Prompt. This is easy to do if you go to the start menu. Then click on Windows System and then on Command Prompt. Type in the word python. This will help us to get the interpreter of Python up and running and ready to go for us.

When we have the steps above done, which will really only take a few minutes overall and won't be as bad as it may seem when you first get started, you will be able to use the Python program on your Windows system. You can then choose to open up the interpreter of Python and anything else that you need, and writing out all of the codes that you need in no time.

Installing Python on a Linux Operating System

And finally, we need to take a moment to see some of the steps that are needed to make sure that we can properly install the Python language on our Linux operating system. Linux is not quite as big of a household name as we will find with the other two operating systems that we spent some time within this guidebook. But there are so many uses for it, and it is able to provide us with a bunch of neat features and more. There are many programmers who already use this language, and many more who are likely to do it in the future, so it is a good thing to know a little bit more about when it is time to install our Python coding language on it.

The first step that we need to take is to see whether or not there is already a version of Python 3, or really even any version of Python found on our operating system yet. The best way that we can get all of this done is to open up the Linux command prompt, and then write out the following code to see what information we are able to get:

- `$ python3 --version`

If you are on Ubuntu 16.10 or newer, then it is a simple process to install Python 3.6. you just need to use the following commands:

- `$ sudo apt-get update`
- `$ sudo apt-get install Python3.6`

If you are relying on an older version of Ubuntu or another version, then you may want to work with the deadsnakes PPA, or another tool, to help you download the Python 3.6 version. The code that you need to do this includes:

- `$ sudo apt-get install software-properties-common`
- `$ sudo add-apt repository ppa:deadsnakes/ppa`
- `# sudo apt-get update`
- `$ sudo apt-get install python3.6`

The good news here is that if you are working with other distributions of Linux, it is likely that you already have Python 3 installed on the system. If not, you can use the distribution's package manager. And if the package of Python 3 is not recent enough, or not the right one for you, you can go through and use these same steps to help install a more recent version of Python on your computer as needed.

And there we have it! We have now gone through and downloaded Python onto the three major operating systems that are out there. this is going to make things so much easier when you are ready to handle some of the codings that you want. And if you used the codes that we described, and went through the other steps that we talked about, you will find that working with the Python language is going to be easy. You can now just open up the right files and all of the parts that you need, and it will work well for you!

Chapter 3: The Basics of the Python Code



At this point, we are going to know a bit about the Python code and why it is so important to helping you get a lot of projects done. We also know the steps that are needed in order to download and install this code, and all of the necessary files, on your computer. Now it is time for us to move on to some of the basic parts that come with the Python code and how we are able to use these to help us go through all of the more advanced codes that we will need to do later on.

The Keywords

The first thing that we need to take a look at is the Python keywords. These are basically going to be the words that we need to reserve. These are going to be the commands that you will need to give the compiler when you would like it to do. We need to make sure that we are using the keywords in the proper manner and ensure that we are not using them in the wrong place, or the compiler will really get confused and raise an error.

There are a number of keywords that are already found in the Python code, and it is important to use them in the right place when we are working on some of the codes that we want to do with this program. Remember that the compiler is using these as commands to tell it what to do. So, treat them as something special along the way as well.

Looking at the Comments

The second thing that we need to take a look at when it is time to start writing some codes in the Python language is working on the comments. These comments are going to be important when it is time to leave a little note or a message inside of your code. This is helpful when you would like to leave this message, and you want to make sure that it is not going to interfere or mess with any of the rest of the code that you are writing.

To make this happen in Python, we just need to add the `#` symbol before any of the messages that we would like to use. You are able to add in as many of these as you would like, but you need to make sure that you keep it as limited as possible. This will help us to keep some of the coding as nice and neat as possible and will make it easier for no problems to show up because of the coding that you are trying to complete.

These comments are a nice way to work with the code and leave the messages and more that you are looking for. This will make it easier for you to write out something that you need to remember or that you would like another coder to know about before getting started, and all you need to do is add in the right symbol in front of the work that you are writing.

The Importance of Variables

The Python variables are also going to be important when it is time to handle some of our codes. And in specific, we need to spend some time learning how to assign a value over to the variable that we are working with. The variable is just going to be a blank piece of the memory if we don't make sure that some kind of value is assigned over to it. This means that we will not be able to get something to show up when we try to pull that part of the code. If the variable assigns some value to it, then it is going to be able to react in this manner when you run the code.

As we work with some of these variables in our code, you may find that there are going to be three main options that you have available to use. Each of these is going to be useful, and it will depend on the type of code that you plan to use and the value type that you want to assign to that variable. The

main variables that you are able to work with will include:

- Float: this would include numbers like 3.14 and so on.
- String: this is going to be like a statement where you could write out something like “Thank you for visiting my page!” or another similar phrase.
- Whole number: this would be any of the other numbers that you would use that do not have a decimal point.

As we do some work with these variables, we have to remember that we do not have to spend our time creating some declaration to save up this particular spot in memory. This is needed in some of the other coding options that you use, but it is going to happen automatically when working in Python right when you assign a value over to the variable with the help of the equal sign. If you would like to make sure that this is going to happen, you are able to look to see that this equal sign is there, and then you know that it is going to work the way that you want.

The good news here is that assigning a value over to the variable is going to be pretty easy. Some of the ways that we are able to do this inside of our code will include:

```
x = 12                #this is an example of an integer assignment
pi = 3.14             #this is an example of a floating-point assignment
customer name = John Doe #this is an example of a string assignment
```

Another option that you can do here, and that we have mentioned a little bit in this chapter already has a variable assigned to two or more values. There are some instances when you are writing code, and you will need to take two values and place them with the same variable.

To do this, you simply need to use the same procedure that we talked about above. Just make sure that you add in an equal sign to each part to help the compiler know that these are all associated with the same variable. So, you would want to write out something like `a = b = c = 1` to show the compiler that each of these variables equals one. Or you could have `1 = b = 2` to show that there are two values to a variable.

The most important thing to remember here is that the variable needs to be assigned to a value in order to work in the code and that these variables are simply spots that are reserved in the memory for those values. Then, when the compiler needs to call up those values, it simply will call up the variable to bring everything out.

Bringing in Python Strings

For some of the codes that we are going to write out in Python, there is going to be a type of data that is known as a string. These strings are going to just be a series of characters that are found in your code and can help you to get different things done in the process as well.

We also will be able to see that there are a variety of operators that we are able to work with when we want to make sure that the strings are going to work with our Python code. An operator is going to just be a symbol that is able to perform a specific operation in the code that you are writing. In this case, the operation is just going to be done on the string and nothing else. Some of the operators that are going to be nice for helping us to get this done and to ensure that the code will behave will include:

1. The concatenation operator: This is going to be the operator that will come in use when you would like to concatenate some of your strings.
2. Repetition operator: This is going to be the best operator to use when it is time to create many copies of the same string. You will be able to make some decisions on how many different times you would like to repeat that string and how many copies you would like.
3. The slice operator: This is the best operator for a programmer to use when they want to be able to go through a specific string and then pull out a character that they would like. Any time that we are working with this option, we are going to have to remember that 0 is going to be the first character of the string, and then you can move on from there.
4. Range slice operator: This is a good operator that we are able to

work with when we wish to a range of characters from the index, rather than just picking out one character. This would work if you would like to show off one word, one part of the sentence, or one sentence.

5. In operator: This operator is going to search for a specified character in a target string. If the character is present somewhere in the string, then you will get an answer of True returned to you. If that character is not inside the string, then you will get an answer of False returned to you.
6. No in operator: This is the operator that will work in the opposite manner as the in operator. It is going to search for a specified character in your string. But if the operator is not able to find that character in the string, then you will get the True answer returned. If that character is found in the string, then it is going to return False.

The Python Functions

Another part of our basics that we are going to look at is the Python functions. These functions are just going to be a set of expressions that we will know as statements as well. They are going to be some of the first objects that fit into classes, which means that you will not have to concentrate on a ton of restrictions as well. There will be a lot of freedom on what you are able to use. You will instead be able to use these functions how you would like, and they are going to work in a manner that is similar to things like strings, numbers, and more. And these are going to also come with some attributes that we are able to bring out as long as we are working with the prefix of `dir` for our function.

One thing that you are going to find is that with these functions, you are going to find a ton of diversity that comes with them. And there are also a lot of attributes that are there for you to use to create and bring up these functions. Some of the different choices that you are going to see with these functions will include:

- `__doc__`: This is going to return the docstring of the function that

you are requesting.

- `Func_default`: This one is going to return a tuple of the values of your default argument.
- `Func_globals`: This one will return a reference that points to the dictionary holding the global variables for that function.
- `Func_dict`: This one is responsible for returning the namespace that will support the attributes for all your arbitrary functions.
- `Func_closure`: This will return to you a tuple of all the cells that hold the bindings for the free variables inside of the function.

There are different things that you can do with your functions, such as passing it as an argument over to one of your other functions if you need it. Any function that is able to take on a new one as the argument will be considered the higher-order function in the code. These are good to learn because they are important to your code as you move.

The Operators

We can also spend a little bit of time taking a look at the operators that show up in the Python language. These operators are going to help us to make our codes a little bit stronger and will ensure that we are able to see some great results in the codes as well. We will also find that there are many types of operators that we are able to work with, based on the operator that you want to work with.

For example, there are arithmetic operators. These are the ones that allow you to add, subtract, multiply, and divide parts of your code. There are comparison operators that will allow us to have a way to compare one or more parts of the code with each other. And we can even do assignment operators that are going to help us to work on putting a value to the variable. You have to choose which option is the best for the coding that you do.

There are a lot of different parts that you are able to work with when it comes to some of the basics of the Python code. Knowing how to make some of these parts work for your needs and the codes that you want to write is going to be important. Even some of the more complex codes that you will work on later, you will find that working with some of these basics is going to help us

to get through that part of the code.

Chapter 4: The Different Data Types in Python



Data is going to be a really important part of the Python language. No matter what kind of coding you are trying to work on at the time, you will find that Python is going to be able to take control over it, and ensure that you are able to see some amazing results in the long run. So, with that in mind, we need to spend some of our time learning a bit more about the different data types that are found in Python, and how we are able to use these for our needs.

For the most part, the types of data will be able to help define the format, will set up the upper and lower bounds of the data to help the program use it appropriately and more. But we will find that the data types that we use in Python are going to be a bit more than that. In Python, we don't even need to spend time declaring the variable by explicitly mentioning the data type. This is known as dynamic typing and can really save us a lot of time and hassle in the process.

Python is able to determine the type of a literal just by looking at the syntax

that is found at runtime. For example, the quotes are going to be there to mark the declaration that is found in a string value, the square brackets will show us a list, and the curly brackets are going to be for the dictionary. Another example of this is when the decimals that are not numbers are going to be assigned to a type of integer, but then the ones that have a decimal in them will be seen as a floating number instead.

Everything, including the modules, the functions, and variables will be a Python object. Another interesting fact here that we can look at is that the variables are not going to have types like they do in other languages. Instead, we will find that they are just going to be labeled. It is the value that we are going to associate over with a variable that will have a type. This is why the same variable can talk about different Python data types.

With this in mind, there are quite a few different types of data that we are able to use in the Python language. We are going to take a look at some of the most common ones below:

First, we are going to have a Boolean. This is going to be a data type that is found in almost all of the programming languages that we can work with. The Boolean that we work within Python is going to come with two values, the True and the False. These values are going to be constants, and we are able to use them to either compare or assign a Boolean value.

Another type of data that we are able to work with is the numbers. These are actually going to be one of the most prominent of the data types in Python. Unlike what we are going to see with some of the other languages out there, the ones that will rely on just floats and integers, Python is going to introduce a few different parts, such as the complex, to help us handle these numbers. The things that we need to remember when it comes to working with numbers in this language will include:

1. The numbers that we see in Python are going to be classified with some of the keywords that you may have seen already, including complex, float, and int.
2. Python is going to have a function known as `type()` that is already built-in, and that will determine the data type that comes with a variable or with a value we are using.

3. Another function `instance()` is going to be built in and will help us to test for the type that we have with an object.
4. We are also able to work with imaginary and complex numbers when it comes to working in Python. We just need to add in the “j” or “J” to help with this.
5. To help us to form a complex number, we are able to use the type as a constructor if we would like.
6. The integers in this language are not going to come with a size limitation at all, as long as we already have some of the required memory in place.
7. A float type number is able to have some precision. This can’t go on indefinitely, though. But we can see it go up to 15 decimal places if we would like.

Another data type is the string that we talked about a bit before. This is going to be a sequence that has one or more characters that we are able to enclose within one single quote or in a double quote. We are able to put any kind of number, letter, or symbol into the string that we want to work with. You will find that the Python language is also able to support a multi-line string. You just need to make sure that there is a triple quotation mark right at the beginning of this string and then one at the end.

Next on the list is the byte. This is going to be considered a data type that is immutable. It is able to store a sequence of bytes, which is each going to be 8-bits, which will range from between 0 to 255. Similar to what we see with an array, it is going to be able to fetch the value of one byte when we use the index. But we will not be able to modify the value that we are working with.

Now that we compared the byte to a string a little bit, we need to make sure that we know how they are different. Some of the main differences that come with these include:

1. The byte objects are going to hold onto a sequence of bytes. But the string is going to be responsible for storing a sequence of characters.
2. The bytes are going to be an object that can be read by a machine. But the strings are going to be in a form that humans are able to read.
3. Since the byte is something that the machine is able to read, it can

be stored right onto a disk without more work. But the strings have to go through and have some encoding done before we can change them to be on a disk.

One of the situations where we are going to find that these bytes really do matter is when we are carrying out the I/O operations with buffering enabled. We may have, for example, a program that is always receiving all of the data that comes over the network. This program is going to parse the data after waiting for the terminators of the headers of the message to appear in the message. It is going to keep appending the incoming bytes to a buffer.

Now we are able to move on to the lists. The Python list is going to be a construct that is similar to an array, and it is able to store some of the arbitrarily typed objects in an ordered sequence that you would like the most. It is going to be a very flexible option when it comes to working with data types, and it doesn't have to come in at a fixed size. The index is a list that is going to begin on 0 when we are in Python.

Another definition that we are going to see when we are working with lists in this language is that these are going to be a heterogeneous collection of items of varied data types that we are able to work with. For example, when we have a list object, we are able to store the files onto a folder, or the employee data in a company and know where it is later on. These lists and the objects that are found in them are going to be mutable. Python is going to allow us to make some modifications to a list, and any of the elements that are in it through assignments and we can work with some of the list methods that come with Python as well.

We are also going to spend some time looking at the Tuples in this language. A tuple is a bit different, but there may be some similarities that we need to focus on. A tuple is basically going to be a heterogeneous collection of objects in Python that are separated out with commas. This means that the objects of different types of data are able to be together in one tuple. The tuple and the list are going to be similar because there are a lot of traits that they share together, including the following:

1. Both of the objects are in an ordered sequence.
2. They are able to store values of different types

3. Nesting is something that we are able to do in these.
4. They are going to allow us to work with indexing and repetition as much as we would like.

However, the tuple and the list are going to be different. The biggest reason that they are so different is that the tuple is going to be seen as immutable. Python does not allow us the option to make modifications to the tuple once it is created. This means that we are not able to remove or add any kind of element at a later time. If we do need to go through here and make some changes, then we will need to go through and create a brand new one with an updated sequence of elements to get this done.

At some point, you may find that you need to work with something that is known as a set in Python. This is the one option that is going to support some of the different operations of math that you may need to do like symmetric difference, intersection, and union. If you plan to do some mathematical work with the use of Python, then this is the data type that you want.

To keep it simple, a set is going to be an unordered collection of immutable and unique objects. Its definition is going to have some enclosing braces with it, which allows us to have the items separated out with a comma inside of there. since the set is going to get some of the implementations from the Set in mathematics, it is also going to be able to have more than one occurrence within the same element.

You will find that there are many codes that you can write that will need a set. The set type is going to have a big advantage over some of the lists that you may want to write as well. It is able to implement a highly optimized method that is then responsible for checking whether the container is hosting a specific element or not the mechanism here is based on a data structure that we can call a hash table.

And the final data type that we are going to spend some time on in this chapter is the dictionary. This is going to be an unordered collection in Python that has some key-value pairs we can call it a built-in mapping type in Python where the keys are going to provide us a map to the values that we are working with. These pairs are there in this language because they will help us to have a more intuitive method to store the data that we have.

The dictionary is so important because it is going to come in and help us solve the problem that comes up with efficiently storing a large set of data. We are going to spend some time later in this guidebook talking about data analysis and what it is able to do for your business if you let it. But the fact is that most companies are already gathering a huge amount of data to use for some of their own needs. And the dictionary is the perfect tool to make all of this happen. Python has already set it up so that the dictionary object is going to be optimized as much as possible to retrieve the data that you would like in the process.

As we can see here, there are a lot of different types of data that we are able to use when it comes to working with the Python language. Each of these data types is going to be so important to what you would like to accomplish with this kind of process, and you will find that you will have your choice of them based on what is going on in your code at that time. take some time to look over the different types of data and what they are able to do for you, and then see how valuable they will be to your needs.

Chapter 5: The Python Functions

Another topic that we need to spend some time looking at when it comes to the Python language is that of the functions. Blocks of code or some logic in the code that is able to perform a specific operation that you would like are going to be known as a function in Python. We have already spent a bit of time in the previous chapters taking a look at what these functions are all about and how we are able to get them to work for our needs. But now we need to take a bit closer look at what these functions can do and how they are able to help us get the best results in no time.

The Different Types of Functions

There are going to be two main types of functions that we are able to find in the Python language. Knowing what these are and how we are able to make them work is going to be so important when it is time to work with this language. The two main types of functions that show up here will include:

1. The library functions: These are going to be some of the built-in functions that show up in the python language. We do not need to worry as much about some of the logic that is in the library function because these are the ones that Python is able to handle on its own. You will find that some of these library functions that will be there often in your codes may include examples like `round()` and `print()`.
2. User-defined functions: Instead of relying on some of the built-in functions, this language is going to allow a user to create some of their own functions as well. So, if you would like to do something like performing a mathematical calculation in these codes, then you would be able to place them in a separate function with the right name, and then you are able to call up that function many times.

The Advantages of Python Functions

With that understanding in place, we need to take a look at some of the advantages that we are able to see when we are working with our functions in Python. There are many of these, which is why you are going to see these functions show up in a lot of the work that you do and in many of the codes that are needed for this language.

The first advantage that we are going to see with these functions is that it is able to take some of the larger programs that you are writing out, and then divide them into smaller groups. This may seem like a bad idea, but it is done so that we are able to gain a better understanding of the code in a faster manner, and it will really make the process of debugging the program easier and better in no time.

Another benefit is that some of these Python functions are going to prevent us from writing out the same kind of logic more than one time. This is good news because it helps us to wrap up the logic in just one function, and then we can call it up the same many times over.

Then there are two more benefits that are going to come up when we are able to work with some of the Python functions. First, we will find that these

functions are going to allow more than one person to work on the exact same program at the same time. this is because we can assign different functions to each person on the team and let them get to work on that. It is also able to help us out because we can call up the same function with different inputs over and over again, as many as we would like, and can save a lot of time and hassle in the process as well.

The Syntax of the Function

We also need to spend some time taking a look at the way that this is going to appear in some of the codings that we do. To start, we need to look at the syntax that comes with Python functions, and then we can move on to an explanation of some of the different parts that are in it, so we know more about how this is supposed to work.

```
def Function_Name (Parameters): Local Variable Declaration Programming  
Logic Executable Statement 1 ..... return
```

1. Def: This is going to be the keyword that will introduce the function definition that we are working with. Keep in mind here that the def keyword needs to be followed right away with the Function_Name.
2. Function_Name: This is going to be any name that you would like to give other than the reserved keywords that we talked about before.
3. Parameters: Every function of Python is going to accept any amount of parameters that you would like. You can have none, or you can have many. It is going to be dependent on what the requirements of the user are all about.
4. The local variable declaration: Sometimes we may need to go through and use a temporary variable. These are going to be the kind that is only needed for that function then we can declare them inside of the function that we are working with. It is not something that is mandatory here, and it is going to depend on the kinds of requirements that you want to have for your work. Remember that these variables are going to be available just to one particular

function, and you will not be able to access them when you are outside of that function.

5. Logic: This is going to be anything mathematical or any type of code or calculation that you would like to be able to implement on that kind of function.
6. Executable statement: This is going to be any of the print statements that tell the code to print some data from the function that you would like.
7. Return: This is another keyword that we are able to focus on because it is going to be needed if we would like to return something from our function. For example, we can use this to help us return the sum of our two integers and more.

There are a lot of things that we are able to do when it is time to bring out the functions and get them to work the way that we would like. When we are able to bring out those functions and use them properly in the code, they are going to ensure that we get the best results out of this possible and that our code stays strong and secure as well.

Chapter 6: How to Write Your Own Conditional Statements

The next thing that we are going to take a look at is a conditional statement. These conditional statements are going to make it easier for the system to respond to any kind of answer that the user is able to give, without the programmer having to know what this is ahead of time. It is pretty much impossible for the programmer to go through and know what answers the user is going to provide in all situations, so working with these conditional statements will help the program run in a proper manner to the information that the user is going to provide.

There are a lot of different programs that are going to rely on these conditional statements. These can often be seen as something that is a bit more simplistic to work with, but this is always something that we are able to add to our code and see some great results. The three conditional statements that you are likely to see along here is the if statement, the if else statement, and the elif statement. Let's take a look at each one and who it is able to work.

Starting with the If Statement

The first of these conditional statements that we need to take a look at is the if statement. You will quickly find that this is one of the most basic of the three conditional statements because it is only going to let the program that you have written proceed forward when the user provides the right answer based on your condition. With this one, if the user ends up putting in an answer that is seen as false based on what you have put into your code, then the code is going to stop, and nothing is going to happen. If the answer though, is seen as true on your conditions, then your code is going to display some kind of message on the screen, or there is going to be some other task that you assigned.

As we can see already, there are going to be some times when if the statement

is going to cause problems in the codes that you are writing, but it is still going to be an important thing for a beginner to learn how to use these statements. A good example of the syntax that you are able to write out to handle some of these conditional statements will include:

```
age = int(input("Enter your age:"))  
if (age <=18):  
    print("You are not eligible for voting, try next election!")  
print("Program ends")
```

There are a few things that will show up with this code. If you have a user go to the program and state that their age is under 18, then the program will work and display the message that is listed there. The user can read this message and then end the program right there.

But, things can go wrong if the user puts in that their age is above 18. This is true for the user, but because it doesn't meet the conditions that you wrote out, the program will see it as false. As the code is written right now, nothing is going to happen because it isn't set up to handle this. The user will just see a blank screen any time they enter an age that is over 18.

Moving to the If Else Statements

The second type of conditional statement that we need to take a look at is the if else statement. We can already see that while the if statement is a good one to practice with, but they're probably are not too many times in the program when you would be able to use this kind of statement. It is likely that the codes you are trying to write will benefit if the user is able to work with whatever answer the user puts in, no matter what their age is. And it is not likely that you want to leave your code so that nothing shows up all of the time.

A better option to use is the if else statement. These statements will provide an output to the user, no matter what they provide as their input. With the example above, the user may get the previous message if they state their age is 16, but then the code would also have a response if the user says their age is 32. It will respond to any answer that it is given, helping the code to keep

moving through the program and ensuring that nothing just stops.

With the voting example that we had above, you can implement the following code to make an if else statement:

```
age = int(input("Enter your age:"))
if (age <=18):
    print("You are not eligible for voting, try next election!")
else
    print("Congratulations! You are eligible to vote. Check out your local
polling station to find out more information!")
print("Program ends")
```

With this option, you are adding in the else statement, which will cover every age that doesn't fall under 18. This way, if the user does list that as their age, something will still pull up on the screen for them. This can provide you with more freedom when working on your code, and you can even add in a few more layers to this. If you want to divide it up so that you get four or five age groups and each one gets a different response, you simply need to add in more if statements to make it happen. The else statement is at the end to catch everything else.

There are a lot of options that we are able to use with this one. For example, you can work with the code syntax that we had before and then ask your user what their favorite color is. you can then work with the if else statements to help you cover up some of the basic colors that you would like. If the user is able to put in one of these colors, then the statement that you put in it is going to show up on the screen. The else statement is going to be the part that we can add to hold onto any of the other colors that you did not plan on, and gives us a good statement to go with it as well.

As we can see, there are a ton of different benefits that are going to come with using the if else statement. These are going to come in to help us really get some results, no matter what input we are getting from the user. There are a lot of programs where you will be able to use the if else statements and get the best results.

Finishing Off with the Elif Statements

And the third type of conditional statement that we are going to work with is the elif statement. These elif statements are going to add on another level, but they are going to be easy to work with as well. You are able to add in as many of the elif statements to the code as you would like to help with the code, and this can help us to cover up any of the different conditions or statements that we want to cover in the code. Think of this elif statement in a manner similar to what we see with some old games, the ones where a gamer is able to pick from options on a menu to decide what actions they would take. The elif statement is going to work in a similar manner.

From this, we are going to be able to work with as few or as many options as the elif statement. You are able to go through and just two or three of these, or you are able to expand out and have thirty of these if you want. A good rule here, though, is that less is more in many cases. You don't want to have too many of these in place, or you are going to end up with a mess, and that is a lot of code to write out in the process as well.

Now that we have had a moment to discuss what these elif statements are about, and how they differ a bit from the if statement and the if else statement, let's take a look at some of the syntax that we are able to work on when it comes to programming with these statements:

```
Print("Let's enjoy a Pizza! Ok, let's go inside Pizzahut!")
print("Waiter, Please select Pizza of your choice from the menu")
pizzachoice = int(input("Please enter your choice of Pizza:"))
if pizzachoice == 1:
    print('I want to enjoy a pizza napoletana')
elif pizzachoice == 2:
    print('I want to enjoy a pizza rustica')
elif pizzachoice == 3:
    print('I want to enjoy a pizza capricciosa')
else:
    print("Sorry, I do not want any of the listed pizza's, please bring a
Coca Cola for me.")
```

With this option, the user is able to choose the type of pizza they want to

enjoy, but you can use the same syntax for anything you need in your code. If the user pushes the number 2 in the code, they are going to get a pizza rustica. If they don't like any of the options, then they are telling the program that they just want to have something to drink, in this case, a Coca Cola.

This is a simple way to use the elif statements to give the user a set of choices. In the other options, the user could add in any choice that they wanted, but in the elif statement, they can either pick that they want one of the choices that you provide, or they will have to go with the default option at the end. This can work well for many games that you may want to pick, for some tests online, and other programs where you want to limit the choices that the user gets to pick from in the code.

And this is how the conditional statements are going to work for your needs. You are able to set up any kind of condition that you would like to work with to ensure that the program is going to work the way that you want. You can set it up with one true condition like the if statement, you can set it up so that you can handle all of the answers that the user can give and still get some kind of response out of the process, or you are able to work with the elif statements and do something that is similar to how a menu works in a game. Each of these can be important, and it is easy to add these conditions into our code to add in some power.

Chapter 7: The Python Classes and How to Write Your Own

One thing that is really neat about working with the Python coding language is handling some of the classes that come with this language. These classes are going to be the containers in the language that is able to hold onto all of the objects, or the different parts, that come with some of the code that you are writing. Being able to write these classes and get the right kinds of codes into them is an important part of keeping your code as organized and easy to use in this language as possible.

That is why we are going to spend some time in this chapter taking a look at how we are able to work on creating a new class. This is one of the best ways to organize the code and will make it so that nothing will get lost or move around in the wrong manner when it is time to execute the code that you want to write. To get started with the process of making one of these classes though, we need to make sure that we use the right keywords.

With the classes, you have some freedom in naming them anything that you would like. We just need to make sure that we are not using a keyword as the name, and that the class name is going to show up right after the keyword that we are using. And it is usually a good idea to pick out a name for the class that you will be able to remember and pull up later.

After you have had some time to go through and name your class, we also need to spend some time naming the subclass. This is the part that is found inside of the parenthesis. And then add in a semicolon at the end of this line so that you match up with some of the good coding protocols in Python that are available to make things easier.

With all of this running through your head, it is likely that this process, and the idea of creating your own class, is going to seem really complicated. That is why we need to stop for a few minutes and look at an example of how we would want to write all of this out in Python. The code that we are going to use to help us to write out a class in the Python language will include:

```

class Vehicle(object):
#constructor
def __init__(self, steering, wheels, clutch, breaks, gears):
self._steering = steering
self._wheels = wheels
self._clutch = clutch
self._breaks = breaks
self._gears = gears
#destructor
def __del__(self):
    print("This is destructor....")

#member functions or methods
def Display_Vehicle(self):
    print('Steering:', self._steering)
    print('Wheels:', self._wheels)
    print('Clutch:', self._clutch)
    print('Breaks:', self._breaks)
    print('Gears:', self._gears)
#instantiate a vehicle option
myGenericVehicle = Vehicle('Power Steering', 4, 'Super Clutch', 'Disk
Breaks', 5)

myGenericVehicle.Display_Vehicle()

```

If you would like, you can try out this code. Just open up your text editor and type the code inside. As you work on writing this out, you will notice that a few of the topics we have already discussed in this guidebook show up in this code. Once you have a chance to write out and then execute this code, let's divide it up and see what happened above.

Now the first part here is the class definition. This is the part where we have to instantiate the object, and then we end up with the definition of the class. This is going to make it easier to know the right syntax of the code. During this time, pay some attention to the code because it is the part that will tell the compiler what you would like it to do, and it is going to highlight the commands that you think are the most important there. if you would like to take the time to bring on a new kind of class definition to the code, then you

will be able to work with the functions of `object_method` or `object_attribute` to help.

Then we are able to work on some of the special attributes. These are found in many of the codes that you want to write out, including the one above. These attributes are good because they will provide the programmer with some peace of mind, ensuring that this attribute is going to be used in the proper manner and that you will not mess it up. There are a few options with these that are found in the code that we wrote before, but there are a few other special attributes that you can consider in the codes that you would like including:

`__bases__`: this is considered a tuple that contains any of the superclasses

`__module__`: this is where you are going to find the name of the module, and it will also hold your classes.

`__name__`: this will hold on to the class name.

`__doc__`: this is where you are going to find the reference string inside the document for your class.

`__dict__`: this is going to be the variable for the dict. Inside the class name.

And finally, we need to take a few minutes to look at the steps that are needed in order to access the members of one of your classes here. You want to make sure that while you are writing the code, both the text editor and the compiler are set up to recognize any of the classes that you are writing. This will ensure that your code is going to be executed in the proper manner.

But for the compiler to execute the code in the right manner, then the code is going to need to be written properly. There are going to be a few different methods that the programmer is able to choose to make this work, and all of them can help you get the work done. However, we are going to focus on working with the accessor method because it is a common option to work with, and it is often going to be the most efficient to get the work done.

To help us to understand the process that is needed in order to access the different members of your created class, the first thing that we need to do is take a look at the code below:

```
class Cat(object)
```

```

    itsAge = None
    itsWeight = None
    itsName = None
    #set accessor function use to assign values to the fields or member
vars
    def setItsAge(self, itsAge):
        self.itsAge = itsAge

    def setItsWeight(self, itsWeight):
        self.itsWeight = itsWeight

    def setItsName(self, itsName):
        self.itsName = itsName

    #get accessor function use to return the values from a field
    def getItsAge(self):
        return self.itsAge
    def getItsWeight(self):
        return self.itsWeight

    def getItsName(self):
        return self.itsName

objFrisky = Cat()
objFrisky.setItsAge(5)
objFrisky.setItsWeight(10)
objFrisky.setItsName("Frisky")
print("Cats Name is:", objFrisky.getItsname())
print("Its age is:", objFrisky.getItsAge())
print("Its weight is:", objFrisky.getItsName())

```

Before we move on, type this into your compiler. If you have your compiler run this, you are going to get some results that show up on the screen right away. This will include that the cat's name is Frisky (or you can change the name to something else if you want), that the age is 5, and that the weight is 10. This is the information that was put into the code, so the compiler is going to pull them up to give you the results that you want. You can take some time to add different options into the code and see how it changes over

time.

Keep in mind during this process that there are often a lot of classes that you are able to work with inside of your Python code, and they are meant to make life easier in coding, not more difficult. These classes can come into use when you would like a better way to take care of your information and make sure it is in the right order so that it makes a lot more sense. You are able to create the class that you would like and add in the objects that match the most.

But do keep in mind that when someone else comes and takes a look at some of the code that you are writing, they should be able to look into the class and understand why some of the objects are found there. The objects don't need to be identical with this, but they do need to make sense as to why they are going to be found in the same class.

There are a lot of times when a class is going to be useful inside your code. And it is definitely worth your time to take a look at them and learn how to use them. Most of the codes that you are going to work on and write in Python are going to rely on these classes, so take some time to look through this code and see how you are able to write out some of the classes that you want to do.

Chapter 8: Handling Files in Python

Working with files is going to be a topic that comes up in the Python language. When you are working in Python, there are going to be times when we would like to take some of the data that we are working on and then store it, making it accessible for you to use later when the data is needed. You are able to choose to save this data in a manner that you are able to find it later on and then see it show up at the right time in any code that you are working with.

This data is something that you are able to save, just like a file on a disk. Or you can make the decision to reuse the data as often as you would like, doing something like an inheritance, as long as you make sure that the correct steps are done along the way. While inheritances are an important part of a lot of the code that we want to write, we need to focus more now on how to create and work with some of the files that are found on the Python code.

Now, there are going to be a few things that we are able to work with when it is time to handle the files in Python. To make sure that this is as easy to understand as possible, we can imagine the files as similar to what we are going to see in a Word document that you have written in, and then at some point, you will want to save them. These files are going to work in a similar manner, but rather than saving the pages, you will need to go through and save some of the code that you are writing.

In this chapter, we are going to take a look at a few of the different things that you are able to handle when it is time to work with some of these files in Python. We are going to take a look at how we are able to write out a new code onto a file that we were able to create before. We are going to look at how we can seek out or move a file to a brand-new location. We are going to look at how we are able to create a file. And then we can finish up with some information on closing and saving a file that we worked on.

Creating Our Own New Files

One of the first things that we need to focus on when it comes to working with these files is how to create a new one when you would like so that you can add in some code to that file. If you want to make your own file and create it with some code on it, then it is important that we open it up inside of the IDLE. You will then need to choose the model that you would like to use when you write some new code. As you work with these, there are going to be three main options for the codes, and you can pick the right one based on what your code is doing. The three modes that are present will include appending (a), write(w), and mode(x).

Any time that you want to make some changes to the current file that is opened, you can use the (w) or write option because this is often the easiest one to work with. Any time that you are trying to open up a file and then write a new string in that file, you will work with binary files and will need the write() method. This is going to work well because it ensures that you will get the right characters returned at the end.

The write() function is really easy to use and allows you to make as many changes to the file as you would like. You can add some new information to the file, or you can make some changes to one that you opened up. If you are interested in doing the writing in the code, you can use this example to make things easier:

```
#file handling operations  
#writing to a new file hello.txt  
f = open('hello.txt', 'w', encoding = 'utf-8')  
f.write("Hello Python Developers!")  
f.write("Welcome to Python World")  
f.flush()  
f.close()
```

Before we take the time to go through and learn more, we need to type the code above onto the compiler and let it execute. This code is helpful here because it will ensure that you can create a new file and get all of the information that you want into the right directory. Keep in mind that the default directory is known as the current directory at all. It is always possible to go through and switch out the directory that you want to use to store the

information, but this is something that you should do before writing any code in it because it is much easier to accomplish.

Whatever directory you are in when you get started is the one where you will need to find yourself back in when it is time to search for that file. If you want to find that file in a new directory, then it is time to move it over there before you save. With the option that we wrote out above, when you go to the right directory, and you open it up, you are going to get the message to show up that is there.

At this point, we have taken some time to write out this program, and we may be happy with it. But then there are going to be times when we would like to do some overwriting of the program. This will change things up so that we will end up with a new statement, rather than the one above, that shows up on the screen when we open that file. This is possible to change things up; even on a previously created file, we just need to make a few changes to the syntax that we are working with. A good example of what we need to write out in order to overwrite our previous code will include:

```
#file handling operations  
#writing to a new file hello.txt  
f = open('hello.txt', 'w', encoding = 'utf-8')  
f.write("Hello Python Developers!")  
f.write("Welcome to Python World")  
mylist = ["Apple," "Orange," "Banana"]  
#writelines() is used to write multiple lines into the file  
f.write(mylist)  
f.flush()  
f.close()
```

The example above is a good one to use when you want to make a few changes to a file that you worked on before because you just need to add in one new line. This example wouldn't need to use that third line because it just has some simple words, but you can add in anything that you want to the program, just use the syntax above and change it up for what you need.

Can I Create a Binary File?

We are going to take a little bit of a detour here and explore for a minute what we are able to do with binary files in Python. This is going to be a bit different, but it is simple and can make a difference in some of the codes that we are writing. You can complete this process in Python because you just take the data that you are working with and then make sure that it turns into a sound or an image file, rather than writing it out as a text file like before.

You can change any of the text files that you write out in this language over to a binary file if that is easiest. In fact, it doesn't matter what kind of file you are working with, whether it is a picture, sound, or text file. The most important thing that we have to remember when we are changing things into binary files is that we need to supply our data inside of the object to make it easier to expose a bit later on. We can then use the syntax below to help us get our files turned into binary files:

```
# write binary data to a file  
# writing the file hello.dat write binary mode  
F = open('hello.dat', 'wb')  
# writing as byte strings  
f.write(b" I am writing data in binary file!\n")  
f.write(b" Let's write another list\n")  
f.close()
```

How to Open a File

We have taken a look at how to create a file and how to turn any file that we have over to a binary file. These are both really important topics when we are exploring what can happen with the files in Python. But now we need to take some time to look at the process of opening up our files.

At some point, you will need to close down a file and move on to something else. Then at a later time, you will want to open up one of the files again and see what is in it or get it to execute it. Once you have taken the time to open up that file, it is going to be easier to work with again. The code that we are able to work with here will include:

```
# read binary data to a file  
#writing the file hello.dat write append binary mode  
  
with open("hello.dat", 'rb') as f:  
    data = f.read()  
    text = data.decode('utf-8')  
print(text)
```

the output that you would get from putting this into the system would be like the following:

```
Hello, world!  
This is a demo using with  
This file contains three lines  
Hello world  
This is a demo using with  
  
This file contains three lines .
```

As we can see here, this is going to be a really simple way that we are able to open up the file that we are working on, and then we can go through and overwrite it as we talked about before, or just write something brand new as a continuation of what we were talking about before.

How to Seek One of the Files

As we go through this chapter, you shall find that these files are a great way to work with and keep the Python script as clean and organized as possible. We have taken some time to really explore a few of the things that we are able to work with when it is time to see results with these files. All of these things can be useful to ensure that the files are going to be opened, written in, and saved in the manner that we would like along the way.

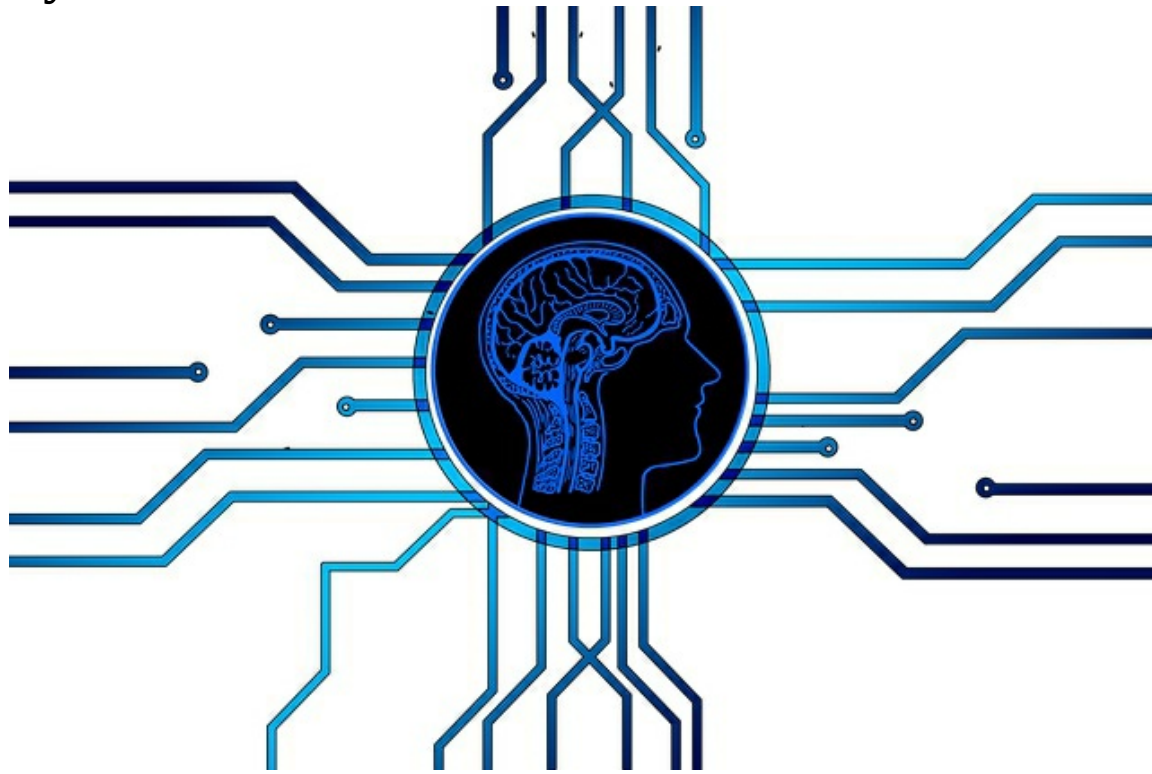
We have taken the time to write out a new file, how to overwrite some of the data that we already wrote in, and even how to open up the files. There may also be some times when you want to move a file to a new place. For example, if you are working on your file, and you find that things are

matching up the way that you want, such as misspelling the name of your identifier the wrong way, or you place it in the wrong directory, you may want to use the seek option to help you find that file and then make the changes.

You will be able to go through and change up the position where the file is so that it goes to the right spot, or even so that it becomes a lot easier for it to find. You just need to tell your code where you would like to find the code and then make the changes that are needed.

Working with files in the Python language will help you out a lot when you are trying to get things in order inside your code when you want to make changes to what you wrote, and so much more. Try out a few of the codes that are inside this guidebook and see just how easy it can be to create a new file, make changes, and get the files to work the way that you would like.

Chapter 9: Tips and Tricks to Get the Most Out of Python



Now that we have had some time to talk about the Python code and all of the different things that we are able to do with that kind of code, it is now time for us to work on how to get the most out of this code. In this chapter, we are going to really focus on some of the things that you can do when your code is not behaving the way that you want, or to help make coding, in general, a bit easier as well. Some of the tips that you as a beginner can follow when working with Python in this manner include:

Comment Out the Code

Every type of coding language that you will work on is going to have the comment, which is basically a way that you can leave a note in your code without the compiler going through and trying to execute these notes as part of the program. This is advantageous to you because you can choose to comment out of a code that you don't want to run right now, but which you

don't want to lose track of. You just need to put the # in front of that line that you want to comment out of.

If your script is long, you can comment on some parts of the code that are not related to the changes that you want to work on. This can sometimes make it run faster, and it can make it easier to search through the rest of the code to help find that mistake. You need to be careful when doing this though because it won't help you to comment out of parts that set the variables that the program needs to use later on. If you do this, then you are going to have issues getting the code to run.

When you are done testing out the code, and you have gotten it all organized and ready to go, you must go back through and remove these comment characters. This helps you to turn the whole program back on so you can see if it works the right way.

Print Things Out

When there are times when you are not able to figure out what is wrong with the code that you are working on, then you can take some time to print out a lot of things to see what is going to work. If you are unsure about which values are ever attached to one of your variables, then print it out. if you are uncertain about anything that is going on in the code that you are writing, then it is time to print it out. Then, when you are actually going through and running the program, you are able to look at your screen and see how the values change or what has been a bit different thanks to that.

You may find that this is going to be helpful to print out some strings before you go through and print out the variable. This is a good way to make sure that your print statements are going to help you to learn how to make the program work for your needs.

Work with the Code You Know Will Behave

When you are in doubt about things, you may want to take some time to practice codes that are in existence already, ones that you know are going to

work, and you don't have to worry about whether they are going to do what you would like or not. This is going to help you to figure out some of the syntax and the structure that has to come with these codes, and then you can work from there to get the best results. This is a good thing for a beginner to just gain some familiarity so that they are able to use that later on as they work on their own codes, or on things that are more complex.

If you do decide to work on your own project, then you can Google around to find a script that has a lot of what you need. It doesn't have to be perfect, but sometimes having that background in place and some of the syntaxes can make a big difference. Make sure that you take the time to run the code before you try to implement any changes. This ensures that the code works before you make any changes. Then you can add in some small changes to that code, and stop to test it often to see if any of the changes that you implement cause the bugs.

Read All of the Error Messages

This one is sometimes hard as a beginner because you may assume that you will have no idea what the error messages mean in the first place, so why should you take the time to learn them and actually read them? But, actually taking your time to read through those and learn how they work, and what they are telling you, can make it easier to fix up some of your code when things are not working well.

You will quickly find that a lot of your error messages are accurate and will actually be pretty descriptive for you. The language runtime tried to execute the program, but it ran into a problem. This would mean that you skipped a step, you had a typo, or something else is missing from your code.

There may be times when you do not understand the message that comes up with an error, but it does try to tell you what went wrong with the code. At a minimum, there will be information on a line number the error is on, and you can head to that part of the code and look for where the bug might be located.

Run the Code Often

Another thing that you need to consider doing when it is time to work with

the Python code is run it on a regular basis. There is nothing worse, especially when you are a beginner, then sitting down and writing code for hours. You may feel accomplished, but when you go and try to run that code, you are going to be in for a surprise. There will be a ton of errors that you need to go through and fix, and then you will feel really frustrated with yourself in the process.

Instead of having hours of code that you need to sort through and try to sort through a bunch of errors, it is best if you try to run the code as often as possible. If you are able to go through and run it every few minutes or after you finish each little part that you are working with, then this is often going to be the best. It will ensure that you are catching some of the mistakes and issues early on in the work, and will make it so much easier for you to really figure out what is going on in the code, and fix the errors.

It is a lot easier to find the errors and the mistakes in a small amount of code than it is to try and do this in a large block of code that you worked on for hours. It may seem like a waste of time and that it is slowing you down to run the code so often. But in the long run, it is going to end up saving you a ton of time and energy and can ensure that you are able to find the solutions that you are looking for. And it makes it so much easier to be able to fix any of the errors that show up because you know exactly where they are going to fall along the way as well.

Take a Break When Needed

Sometimes, working on coding can be a breeze. Everything is going to fit in just as you need it to and you can spend hours working on this part and getting it all to match up the way that you would like. And other times, things are not going to run as smoothly for you and will not work out as well. When the latter starts to happen, it is time to take a break.

Often, we work on the code for far too long. We want to get it done or get something figured out with it, so we will spend hours and hours working on it. And then we get frustrated. Something or other on the code is not working out the way that we want, and we can't seem to get it fixed and working well again. And no matter how hard we try, and how much work we put into it, we

end up just making the whole situation worse in the process.

While it is sometimes hard to just let go of the coding that we are doing and take a break, sometimes this is just what we need. When we are tired and have been staring at something for a long time, it is just going to be worse if we continue to stare at it and try out too many things. When we take a break and learn how to walk away from what we are doing on the computer, it is going to become so much easier.

You may have to force yourself to take this break a bit. When we are frustrated is usually the time when we want to dig our heels into it a bit more, and we don't want to leave all of the work that we are doing at the time. But even an hour break or more can be nice to allow us to get away and rethink things. And when we come back, we can find the mistake or the error or the problem that we couldn't before, and we can get it taken care of.

Ask for Help

As a beginner, there may be some times when you will need help with the coding that you are doing. It is hard for some people to ask for help because they want to be able to do it all on their own. But learning how to code takes some time and patience, and then there are going to be those times when we just need some help because we aren't sure what steps to take next.

When you are asking for help, there are a few steps that you should consider taking in order to make it easier. These help to make sure that the other person knows what you have tried and doesn't waste time going through all of that again as well. Some of the things that you can consider knowing and explaining to the person who is going to help you out with your code will include:

1. Explain what you want the program to do for you and where the error is occurring.
2. Show the other person the code that is sending you an error so they can see it for themselves.
3. Show the other person the stack trace, all of it, including the message you got stating the error.

4. Explain everything that you have already tried on the error. This helps the other person have a good idea of what you have tried and what they should try to get the code to work.

You may find that the process of thinking through these items in your mind and then getting them written down on paper can help make a brand-new solution obvious. If that doesn't work, try to find someone who may be able to look through the code and tell you what is going wrong and what you can do to fix the error.

There are a lot of parts that need to come together when you are working as a programmer for the first time. being able to get all of these to work with one another, and ensuring that you are able to learn how to fix some of the problems that come up in your own code can be so important when you are a beginner trying to figure all of this out. Make sure to check out some of the different steps and tips that you can do when working in Python to make sure that you are coding correctly and that your program or application is going to work for your needs.

Chapter 10: A Quick Introduction to Data Analysis



Many of the topics that we have spent time talking about in this guidebook are going to lead us to a good look at data analysis and what this can entail for us. There are many companies who would like to learn how to work with the Python coding language because they know it is going to help them figure out how to better serve their customers, figure out the best options when it comes to which products to release to their customers, and so much more.

There are a lot of benefits that will come with working on data analysis, but before we do that, we need to first learn a little bit more about data analysis and what this process is all about. Data analysis is simply going to be a process that we can use in order to take our raw data and order and organize it in a meaningful manner. This is going to make sure that we are able to take out some of the important information that is found in that set of data, and then use it for our needs.

The process of organizing and then thinking about our data is going to be so important in helping us figure out what the data is going to contain and what we are able to learn from it. There are going to be methods that we are able to

use in order to approach one of these kinds of analyses for their own needs, but the most important thing that we need to remember here is to keep our biases and any manipulation out of the mix.

It is really easy for someone to get in there and add in some of their own biases and more in the mix. This is sometimes done on purpose if the person wants to make sure some of their ideas are going to be the ones picked. But this is also something that can happen on a more unconscious level as well. No matter where it comes from, it is important to pay attention to the presentation of the data analysis and to really be critical about the process that is used along with it.

Another thing to remember here is that raw data can easily take on a lot of different forms. And we are going to see the data come at us in places like observations, social media, survey responses, and measurements. In the raw form that we usually collect this data from, the information is going to prove to be really useful. But there are also problems with it being overwhelming as well.

Over the course of the process that you take for this data analysis, we need to take all of that raw data, no matter how much of it there is, and then order it in the manner that will make it the most useful for us. For example, we may go through and tally some of the survey results that you are going to see after sending these out to your customers. This will give you a better idea of how many people actually went through and answered the survey, and can help you see how people responded to the different questions that were there.

In the course that you take, in general, to organizing the data, there are going to be some trends that will emerge. And sometimes, we are going to see that these trends are highlighted in the writeup that you see of the data. This is done so that the readers are going to take note. In a causal survey over something, it may be interesting to note that men like one thing better than women, or that women are more favorable to something than men. This is important for marketers to know sometimes, and can make it easier to learn how to market and reach each one.

In addition, modeling the data with the help of things like mathematics and some other tools is going to help us here. These can make sure that the right

points are exaggerated, the ones that are of the most interest to us, which makes them a bit easier for the researcher to see and pay some attention to as they go through the process.

Another thing to add to any data analysis that we are working on is that the charts, textual writeups, and graphs are all going to be important forms of data analysis. These methods are going to be important because they are meant to distill and refine the data in a manner that readers are able to glean the most interesting parts of the information without having to sort through the data all on their own.

Summarizing the data and what all has happened during the analysis is often going to be pretty important when you would like to support any of the arguments that you made with that data, as is presenting the data in a manner that is clear and understandable. The raw data is something that you may choose to include in the appendix in some cases, because this allows those who need it, and why maybe using the information to make important decisions, a way to look through the specifics of the data, of the analysis, and more, if they so choose along the way.

With this in mind, it is time for us to take a look at some of the steps that are going to come to us when it is time to work on our data analysis. These steps are going to be pretty similar no matter what kind of project we are performing or working on in data analysis, and we will be able to see the Python language help us get it all done. Some of the different steps that you are able to take to complete a data analysis successfully will include:

Define the Question

In the business or organizational data analysis that you are working on, we need to make sure that we are jumping in on the right kinds of questions. It is never a good idea to start out with the idea that you will just sort through the data and hope that some good insights will pop out at you. There is just too much data to work with any more. And doing this is going to waste a lot of time and energy. You will end up in a rabbit hole if you do this and not be able to get anything done.

The questions that you rely on to get this work done need to be measurable, clear, and concise. You should also make sure that the questions are going to be there to either qualify or take away some of the potential solutions to the opportunity or problem that we are working with.

For example, a good way to start with this is to make sure that you know your clearly defined problem ahead of time. Maybe you are currently working for the government as a contractor, and you find that a lot of your costs are rising. This makes it a lot harder for you to submit contracts that are competitive, and you might lose your work. One of the questions, out of many that you are able to use in order to solve this business problem, could include whether or not you can reduce your staff without compromising on quality.

Pick Out the Important Measuring Options

Once we know what kind of business problem we would like to focus on, it is time for us to step back and work on creating the clear measurements and the priorities that we want to go with them. This is a step that we are able to break down into two parts, which makes it easier for us to work through. These steps are going to include deciding what you need to measure and then deciding which steps we need to take in order to measure it.

So, with the first step, we have to decide what exactly we would like to measure in the first place. Going back to that government contractor that we were looking at, we would want to focus on what kind of data is needed to help answer these key questions. For this one, it would be a good idea to know how many staff members you have, how much they cost you, and the percentage of their time that they are spending right now on functions that are pretty necessary to the business.

In answering these questions, you are going to then come up with some sub-questions as well. For example, do you see that your staff is not being utilized to their full potential, and are there some improvements to the process that you could do to help with this and reduce costs? There are always a lot of options and questions that you are able to work with on this one, and the more you focus on these questions, and the further down you go with them,

the better you will be able to make a decision as well.

And the final thing that we need to look at when it is time to decide what to measure out of your process, you need to make sure that all of the reasonable objections that may come up if you do go with one action over another. For example, if you do decide to cut down on a few staff, how will you be able to respond if there is a big surge in demand for your services.

You need to then move on to the second step that is going to come with this process. Thinking about how you are going to measure out this data is going to be just as important as all of the other steps, especially before you go through the phase for data collection. The reason for this is that the measuring process that you are working with is either going to be able to back up the analysis that you do later on, or it is going to discredit the analysis that you are doing. There are a few different key questions that you are able to work with here, and they will include:

1. What is the time frame that you want to follow? Think about how long you want to search for the information, how much historical data you want to use, and how long you expect these changes to last. If you are just dealing with a temporary decline, such as a seasonal one, maybe finding another way to reduce costs is better than laying people off.
2. What is the unit of measure that you are going to work with? This could be a comparison of Euro and USD for example.
3. What factors do you think should be included in this process. You could check out the cost of your employees with just their annual salary. However, this is not going to provide you with the full picture because you will be able to learn a lot more if you work with the annual salary, and the cost of benefits that you are providing to your employees can add up as well.

It is important to have a good plan in place before you try to do anything with your data. Even collecting it is something that can be done after you have your measurement criteria in place, and you are sure that you know what steps to take next.

Collect the Data

At this point, we have had some time to go through and clearly define the question that we would like to solve for our business. And then, we also took some time to learn which measurement priorities we need to set as well to ensure we get the right kind of data and more. Now it is time to collect the data. As we go through and organize and collect the data, there are going to be a few points that we are able to keep in mind that will make it easier for us to really see some of the results that we want.

First, before we go through and collect the new data, we need to look at the information that we already have. if you already have some of your own sources and databases that have been collecting data, then you need to look at those first. This can help you to really take a look at this data and save yourself a lot of time and money in the process. Depending on how much data is collected, you may have all of the data that you need without having to look in other locations.

Then we need to move on and determine how we are going to handle the data that we bring in. if we look at some of our own resources and find that we need some more information, we then need to think of a good naming and file storing system ahead of time. This is going to make it easier for those who are on the team to work with this to collaborate with one another. This process is going to save your team some time and can prevent the members from going through and collecting the same information more than one time.

If you end up needing to go through the process of gathering up the data and using interviews and observations to finish this up, then you need to make sure that there is a good interview template that everyone is able to use. This is going to make sure that there is a level of consistency in what you will see, and can save some time.

And finally, we are going to make sure that we keep all of the data that we are collecting through this process as organized as possible. The method you use is going to depend on the data that you are using and how you plan to use it for your needs. You could do something like having a log with the collection dates, and then add in any of the sources notes that you need to keep this all in line. this practice is important to spend some time on because

it is going to help validate some of the conclusions that you are making later on.

Collecting the data that you need to finish this process will help us to really get the data analysis done. You need to make sure that when you collect this data, you are picking it out from some reputable and good sources along the way. the higher the quality that you can put with the data that you collect, the better it is going to be later on. When you perform the analysis, this data is going to give you some of the results and predictions that you are going to put your decisions on, so being a bit selective with this can help.

Look Through and Analyze the Data

And this brings us to our fourth step, which is where we are going to take all of that data that we have been collecting so far, and we will analyze it. After we have been able to collect the data that is able to help us answer the question that we came up with in the first step, it is time for us to go through a deeper analysis of the data. There are a few options and steps that we are able to follow to make this happen, and we are going to spend some time discussing those now.

Begin this with manipulating the data such as plotting out that data and seeing what kind of correlation is there. or you can even create your own pivot table to work within Excel. This pivot table can work for you because it will allow you to do some sorting and filtering of your data by some different variables. It is also going to make it easier to figure out the maximum, mean, standard deviation, and a minimum of the data that you have.

As you are working on manipulating the data, you will sometimes find that you already have the exact data that you need right from the start. But in most cases, you will still need to either gather more data to help or make some revisions to the original question that you were asking. Either way, this initial analysis of the correlations, outliers, variations, and trends will make sure that you are able to focus the analysis that you are doing on answering the question that you originally posted. It is also a good way for us to handle some of the objections that others may have worked raised against you.

During this specific step, the tools and software that come with data analysis

are going to be really helpful. There are a few different advanced processes that you are able to choose from that will help us handle these advanced statistical data analyses that we want to complete. In most cases, though, you are able to keep things simple because you can work with Microsoft Excel and get a lot of the same decision-making tools that you need. There are a lot of features and functions that come with Excel, and you will find that you are able to use this to help you get the best results as well.

The analysis that we do during this step is going to take some time. Picking out the tools that we want to use, and the right algorithm is going to make it a lot easier. We will talk about some of the algorithms that are available to help us complete this analysis and often it is going to depend on how much data you want to work with, and what answers you are hoping to get out of it as well. There are also a few ways that we need to go through and separate our data.

These two methods are going to be the training set and the testing set. Before we are able to get through and actually work on the analysis, we need to train and test the algorithm that we would like to use on the proper methods and what it needs to do overall. The algorithm is not going to come already prepared and ready to go. Instead, we have to show it how to behave and do some work with it, and then we can put through the data we want and get the predictions.

To start, we need to split our data into two sets, the training, and the testing set. The training set needs to be a lot larger than the testing set. This is because we are basically using this as a way to teach the program how we want it to behave, and ensuring that it has plenty of time to see the relationships that are there and figure things out. We need to also make sure that the data we are working with is high quality so that it is going to work for learning faster and helping us later.

After we have had a chance to go through and send the training data into the algorithm, it is then time for us to work with the testing set. We send this one through to see how well the algorithm was able to retain the information that is gathered in the last set. The accuracy the first time is not going to be perfect, but since the algorithm is going to learn more times than it does the process, this is not a big deal. It will get better.

The higher the accuracy that you are able to get, the better. But if you end up getting an accuracy level that is below 50 percent, then this is a sign that something went wrong. Your algorithm should be able to make random guesses on the testing and get 50 percent or more right. So, if you are getting less than this, it is a sign that the algorithm is not set up right, or maybe that the data that you are using in that is not very good, and you need to restart with something else.

You will need to go through the training and testing phases a few times if you are able to. This will make it easier for us to figure out the best way to train the algorithm and ensure that it is set up properly. The more information and data that you are able to send through to the algorithm, the easier things get, and the higher the accuracy will become overall. When you do this a few rounds, you should notice that when you finish with each of the testing grounds, the accuracy rate is going up.

We do not need to have 100 percent accuracy with our algorithm, but the higher that we are able to get this, the better. When we get to an accuracy level that we are comfortable with, we can then move on to actually using this algorithm for some of the information that we need. It is then ready for us to put the data we want into it and learn how to make our decisions and have some good predictions in the process.

Interpret the Results

The final thing that we need to take a look at here is interpreting the results that we get out of the analysis. After we have spent some time analyzing the data, and maybe doing some more research if it is needed, it is time for us to go through and interpret the results that we get from this. As you spend some time interpreting the analysis, keep in mind that you will need to go through and see whether the original hypotheses that you formed was the right one or not. Sometimes it is going to be right, and other times it is going to be wrong. And when the hypotheses are wrong, you need to go through and make some adjustments to figure out how to get things back on track.

As we are going through and interpreting some of the results that we are able to get out of the data, there are a few questions that we need to ask for the

data. We need to first start out by asking whether the data is able to answer our original questions. Think back to that question that we had in the first section of this, and then determine whether the analysis that we just went through was able to answer the original question and how it was able to help out with this.

The next thing that we need to look at is whether the data is going to be strong enough and sound enough to help us defend against any of those objections that we had before. Think back to all of the objections that you had heard when we got started with the project, and think of what you are able to do in order to allay any fears and worries that are going to come up with this.

We also need to make sure that this is the right conclusion that we came up with and that it is going to be the right one for us to work with. For example, we need to take a look and see if there are any limitations on the conclusions that we have, and any other angles that we may not have looked at yet concerning this. The more in-depth that we are able to get with this, the better it is going to be for us when we finally do come up with the conclusion that we want to use.

If you find that the interpretation that you are able to get of this data after finishing the analysis is able to hold up under the questions that we just did and more, and they meet with the considerations that we just had, then you likely have a conclusion that is productive and that you are able to use. The only step that is left for this one is to use the results of the analysis in order to figure out the best course of action to work with.

This is a process that takes some time to accomplish and is not always as easy and straightforward as we may like. It is supposed to be this way. The algorithms are meant to come into this process and speed it up, though, much more than we would be able to see if we tried to go through this process in a manual manner.

When we take the time to follow the five steps that we listed out above to help with data analysis and all of the parts that come with this process, it is easy to see how we are able to make better decisions for your business or any company you are working with. The reason that your decisions are going to be stronger and smarter than ever before is that they are going to be backed

by data that has been collected robustly and analyzed so you know you are able to trust them.

With some practice, and with some careful consideration of what is found in the data that you are looking at, you will find that your analysis of the data is going to only get faster and more accurate. And this is going to mean a lot of good things for your business because it means that you are able to make better and more informed decisions that will help your business run in the most effective manner possible.

Chapter 11: Some of the Best Python Algorithms for Data Analysis

We have spent some time in this guidebook looking at some of the basics of the Python language, as well as some of the basics of data analysis and what we are able to do with this for some of our own needs. Now it is time for us to take both of these ideas and put them together.

In order to complete a good data analysis in the first place, we need to be able to run the data that we have collected through a number of algorithms, usually, a few, though in some cases, one will be fine. These algorithms are going to help us to learn what patterns and insights are found in that data, and can give us some good predictions in the end.

There are a number of different Python data analysis algorithms that we are able to work with along the way, and it is going to often depend on the what we would like to do with that data to start with, and how we are able to benefit from this as well. With this in mind, let's dive right into some of the basics that come with these algorithms, and learn a bit more about why Python algorithms are the best for data analysis.

Neural Networks

The first kind of Python algorithm that we are able to use for data analysis is going to be known as the neural network. This is a fairly sophisticated type of algorithm to work with in the first place, but it is going to provide you with a lot of the power and more that you need to get some of the work done.

The neural network is going to work in a manner similar to how the brain will function. It is going to form connections to the things that it gets right, and it can remember this for future use as well. This helps it to learn as time goes on, and makes it easier and faster and more efficient at some of the work that it needs to complete as well. Once it has been able to go through and learn something new, you will find that it will remember it in the future.

In addition, the neural network is really good at helping with things like identifying what is in an image. It does this by going through the different layers of the pictures, one at a time, and then making a good prediction of what is going to be found in that image. The more layers that it is able to go through, the more accurate it will be at guessing what is in the picture as well. This makes it an effective tool to use for many of the projects you are working with.

However, we have to keep in mind that the computational costs are going to be a bit higher with this one. Because we are working with an algorithm that is able to methodically go through a bunch of layers and make accurate predictions, we can see why the costs will be higher with this one. You have to determine ahead of time if this is the right kind of process for you to work with, or if one of the other algorithms out there would work better.

Clustering

There are a number of different clustering techniques that you are able to work with when it comes to handling a data analysis. These are all going to take the data, though and put them together in clusters so you can see more about the distribution of where your points of data are going to fall. There are different ways that you are able to work with clustering, and different ways that you can cluster up your data, but it is a good way to put the information into groups and see some good insights right away.

The number of clusters that you will have in your work is going to depend on what you are focusing on. For example, if you are working to divide your group into males and females, then you will only need to have two clusters. But if you would like to divide things up by the age of your customers, or by where they live in the country or the world, then you will probably need to have more than two clusters in order to have the best idea of what is going on.

With these clusters, you will be able to plot them on something like a scatterplot, and then figure out where everything is going to fall. This is a great way to figure out the best options to take, whether there is a new demographic that you need to work with, and more. You can just put these

clusters on your scatterplot, and you can learn a lot of information in a short amount of time.

The idea of these clusters is pretty simple. When you have some data points that fall into the same cluster, you can safely make the assumption that the points are going to have some similarities with one another. But if the data points are not in the same cluster, then this is a sign that they are going to be different from one another. If we are able to keep this in mind, it is a lot easier for us to work with understanding this kind of algorithm.

Support Vector Machines

These are going to be the algorithms that you are able to use in order to figure out the right course of action to take. When you see that the line and the hyperplane come together, then this is a good idea that the information that is there is going to lead you to the results that you are looking for. This can make picking out the right decision easier, as long as you know the right steps that are going to help you to get the best results as well.

The biggest trick that comes with this one sometimes is figuring out which lines and hyperplanes we should pick and work with. Sometimes, depending on the type of data that you are working with and what is all in it, you will end up with more than one hyperplane. Sometimes this will not be a big deal because you will be able to see which one is the best for your needs. Other times it may seem like there is more than one answer that is going to work the best for your needs as well.

There are a number of options that we are able to work in order to pick out the right SVM for what we want to do. You will learn how to do this more with experience, but you will be able to learn about your data and then figure this out as well. Sometimes experimenting with a few of these to see which one is the best is another good option that we are able to choose as well.

Naïve Bayes

The Naïve Bayes algorithm is a good choice to go with when you want to do

a bit of exploration with the data in the beginning. Maybe you want to see what the best way is to split up the data that you have, or you are not yet certain about what kind of algorithm is going to be the best one for you to focus your attention on yet. In some cases, you may need to show some of the data and some of the information that you have ahead of time, right after collecting it, to those who want to see what is going on, but may not understand all of the more technical aspects that come with it.

This is where the Naïve Bayes algorithm is going to come into play and can really help us. With this option, we are able to take a good exploration of the data that we have, and then determine the best steps to take after. Sometimes this helps us to choose which of the other algorithms are the best ones for us to go with. And other times, it may be a good way to create a beginner's model so that we can show this off before being able to finish all of the work for the final project.

The Naïve Bayes algorithm is usually not going to be the first choice that we make when it is time to handle some of our data, and we will usually go through and make a few other adjustments to the process as well and finish off with another kind of algorithm. But it is definitely a good algorithm to go with because it adds in a lot of the different parts that you need to get a good idea about what the data contains, and what else we are able to do with it along the way.

Decision Trees

Another option that we are able to go with here is the decision trees. These trees are going to help us to make some good decisions and compare some of the likely outcomes of the decisions that we do make. if you are uncertain about which steps to take, especially if you have a few choices to pick from, then the decision tree is going to be one of the best algorithms for you to choose from.

To start, you would be able to ask your question and implement it in the information or the data that you have. Of course, we first need to clean and organize it, but we will assume that this part is already done. When you are ready, the data can tell us which steps are the best, and the likely outcome of

each choice that you try to make along the way. This can help you to make some better decisions because you can look at the data in a more clear and organized manner, and you know that it actually makes sound sense, rather than just using your intuition along the way.

In addition, you are able to take a bunch of these decision trees and combine them together into a random forest. The random forest is simply going to be when we combine together a bunch of those decision trees in order to compare more than one option. Maybe you want to see which is best for you. If you ever want to compare more than one idea or thought for you to take, then working with the random forest, or even a simple decision tree, can get this done. You just add in the information that you want to work with, and then compare the decisions and their potential outcomes before going.

As we can see, there are a lot of different types of algorithms that we are able to use when it comes to handling our data and making sure that it makes as much sense as possible. A bit point of working with data analysis is to help us to sort through all of that data. Otherwise, that data is just going to sit around and be worthless to us and just sitting there taking up space.

Each of the algorithms that we went through above is meant to help us to figure out different ways that we are able to move around our data, and different ways that we will come to a better understanding of what is inside of our data, and how we can use that. Whether you want to be able to make smarter decisions for your business, or you are looking for ways to find new customers or beat out the competition, the algorithms above, with the help of the Python code, will be able to help you get there.

Conclusion

Thank you for making it through to the end of *Python Programming* , let's hope it was informative and able to provide you with all of the tools you need to achieve your goals whatever they may be.

The next step is to get started with the basics of programming in Python. You will find that even as a beginner, or someone who has never done any kind of coding in the past, working with the Python language is easy enough to learn. And we spent some time in this guidebook learning more about how this is going to work, and how we are able to code in no time.

There are a lot of reasons why people and businesses are going to want to learn how to work with the Python language, and with some of the codes and more that we looked at in this guidebook will show us just how easy working with this kind of language can be overall. It is one of the best coding languages for basic coding all the way up to data analysis and machine learning, so there is always going to be a lot of things that you are able to do with it.

This guidebook took some time to explore more about what Python is able to do for us. We looked at some of the basics, like how to install this language on your own computer, and then moved on to some of the different codes that you are able to write out with the help of this language. And then, we moved on to how we can use this in a more practical manner with the help of a good data analysis to push it along and actually help your business learn from the data and everything it has been able to do overtime.

When it is time for us to explore more about Python programming and what we are able to do with some simple coding along the way, make sure to read through this guidebook and learn how to get started.